

" V is a vector space over $\mathbb{R}$ " means

① V is an 'abelian group' with $V \times V \xrightarrow{+} V$ (assoc, comm.), neutral elt 0 , inverses

② There is a scalar mult. map $\mathbb{R} \times V \rightarrow V$ satisfying 4 properties

$$a \cdot (v+w) = av + aw, \quad (a+b) \cdot v = a \cdot v + b \cdot v, \quad (ab) \cdot v = a \cdot (b \cdot v), \quad 1 \cdot v = v$$

Formal consequences: 0 is unique, inverse is unique, $0 \cdot v = \bar{0}$, $a \cdot \bar{0} = \bar{0}$, $(-1) \cdot v = -v$ Tutorial

Words with same defⁿ as we had for $\mathbb{R}^n$: linear combination, span,

linear (in)dependence, basis for V , subspace of V , linear map $V \xrightarrow{f} W$

Observations about span, independence Tutorial

$B \subset V$ is a basis for $V \stackrel{\text{def}}{\iff} B$ is indep & $\text{span}(B) = V$

We will stick to "f.d.v-sp"

V i.e. V has a finite spanning set, say S

$\iff B$ is a minimal spanning set for V

$\iff B$ is a max'l indep set in V .

$\iff$ Every vector $v \in V$ can be written uniquely as a lin comb. of vectors in B .

This gives "coordinates of v w.r.t. B "

Thm ① Every v-space has a basis ② Any two bases have the same cardinality. We'll prove this for f.dim v-sp.

Pf. ① Start with S . If indep, stop. Else find $v \in S$ $v \in \text{Span}(S \setminus \{v\})$. Now $\text{Span}(S \setminus \{v\}) = \text{Span}(S)$

Repeat with $S \setminus \{v\}$ ② ! We proved this for $V = \mathbb{R}^n$.

Example Suppose $\{v_1, v_2\} \subset \{w_1, w_2, w_3\}$ are bases for V .

$$w_1 = a_1 v_1 + a_2 v_2$$

$$w_2 = b_1 v_1 + b_2 v_2$$

$$w_3 = c_1 v_1 + c_2 v_2$$

Write general proof (Ex.)

Another pf by row reducing transpose matrix

$$\begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \\ c_1 & c_2 \end{bmatrix}$$

Observe: the pf only used

$\{v_1, v_2\}$ span V

$\{w_1, w_2, w_3\}$ indep

to derive a contradiction!

Idea: use these to show that w 's are dependent

$$\text{How? } \left(\begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 w_1 + x_2 w_2 + x_3 w_3$$

$\therefore$ There is nontrivial $x \in \mathbb{R}^3 \in A_{2 \times 3}$ st $Ax = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Multiply both sides by $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ on the right. Use associativity to conclude

$$\sum x_i w_i = \vec{0} \rightarrow \leftarrow$$

Upshot. The same pf gives in a f.dim v.space

If S is a finite spanning set & I any indep set

then $|I| \leq |S|$

Cor ① Any two bases have the same cardinality } In a fin dim'l v.space.

② Any lin. indep subset can be completed

to a basis

Cor In an n -dim'l v.space V , { any n lin. indep vectors form a basis.
any n vectors that span V —//—

An isomorphism of vector spaces is a linear map $V \xrightarrow{f} W$ st. that

it has an "inverse linear map" $V \xleftarrow{g} W$

$$\begin{array}{|c|} \hline \downarrow \\ \hline \begin{array}{l} f \circ g = \text{id}_W \\ g \circ f = \text{id}_V \end{array} \\ \hline \end{array}$$

f and g must preserve $+$ & $\cdot$.

If $\{v_1, \dots, v_n\}$ is a basis of V then we have an isom

$$\begin{array}{l} \mathbb{R}^n \rightarrow V \\ \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mapsto \sum_{i=1}^n a_i v_i \end{array}$$

Observe/easy exercise: If f is a bijective linear map then

(f^{-1} exists set-theoretically b/c f is a bijection and)

f^{-1} is automatically linear (hence f and f^{-1} are inverse isomorphisms)